

Team Boundary Activity: Its Mediating Role in the Relationship Between Structural Conditions and Team Innovation

Group & Organization Management

2014, Vol. 39(3) 274–299

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DOI: 10.1177/1059601114525437

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Abstract

The purpose of this study is to broaden our understanding of team boundary activities in promoting team innovation. The model examines the mediating role of boundary activities in the relationship between the structural variables of inter-team goal interdependence and team functional heterogeneity and team innovation. Sixty teams from high-technology companies were studied involving 196 team members and their 60 corresponding team leaders and unit managers to whom the leaders report. Results identified two dimensions of boundary activities: boundary-loosening activities and boundary-tightening activities. Overall, the results of the structural equation model confirmed the mediating role of team boundary activity. The results indicated a positive relationship between the two structural variables and boundary-loosening activities, a negative relationship between functional heterogeneity and boundary-tightening activities, and a positive relationship between the two dimensions of boundary activities and team innovation. Implications, limitations, and directions for further research on team boundary activities are discussed.

Keywords

boundary activity, teams, team innovation, goal interdependence, functional heterogeneity

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With the increase in competition, complexity, and environmental demands, organizations are adopting and improving team-based structures to ensure the most efficient handling of their subtasks and the highest organizational effectiveness (Ancona & Caldwell, 2007; Drach-Zahavy & Somech, 2010; Marrone, 2010). For obvious reasons, studies have typically referred to teams as isolated entities, separate from their organizations; therefore, the key questions have focused on intra-group structures and processes (e.g., Barrick, Bradley, Kristof-Brown, & Colbert, 2007). But, in fact, an organization's teams function interdependently and do not operate in a vacuum (van Knippenberg, 2003). Each organizational team needs information, products or services provided by other teams; team relies on other teams to promote the interests of all and constructively to resolve possible conflicts. To this end, team must conduct boundary activities to preserve, protect, or acquire its resources (van Knippenberg, 2003).

Over the past three decades much theoretical and empirical work has emerged, advancing team boundary activity research (Ancona & Caldwell, 1998; Choi, 2002; Faraj & Yan, 2009; Joshi, Pandey, & Han, 2009; Marrone, 2010; Yan & Louis, 1999), but the impact of such boundary activities is still not fully understood in terms of antecedents or consequences (Faraj & Yan, 2009). First, although several studies have captured the boundary activity phenomenon in the team context, they have focused on how specific members of the team communicate with external agencies to promote team functioning (e.g., Johnson & Chang, 2000), but they have not considered how the team as a unit operates with regard to its boundary. Second, studies focusing on boundary activities as a team construct tend to emphasize the sharp contrast between inward-looking activities intended to tighten the team's boundaries, and outward-looking activities intended to loosen them (Ancona & Caldwell, 1998; Choi, 2002; J. M. Howell & Shea, 2006; Marrone, 2010). However, several scholars have provided compelling arguments as to why the two kinds of boundary activities might not be conflicting, but instead complementary (Campion, Papper, & Medsker, 1996; Drach-Zahavy & Somech, 2010; Yan & Louis, 1999). Leadership researchers, taking a more holistic view of boundary activities, recognize that boundary-tightening and -loosening activities are equally important for promoting team and organizational effectiveness (J. P. Howell & Costley, 2006). They regard the two as synergistic, not supplementary. Third, while many of the earlier studies on boundary activities examined either their antecedents (Ancona & Caldwell, 1998; Joshi et al., 2009) or their consequences (Ancona & Caldwell, 1998), very few researchers have explored the mediating role of boundary activities in relation to specific antecedents and outcomes (Ancona & Caldwell, 1992).

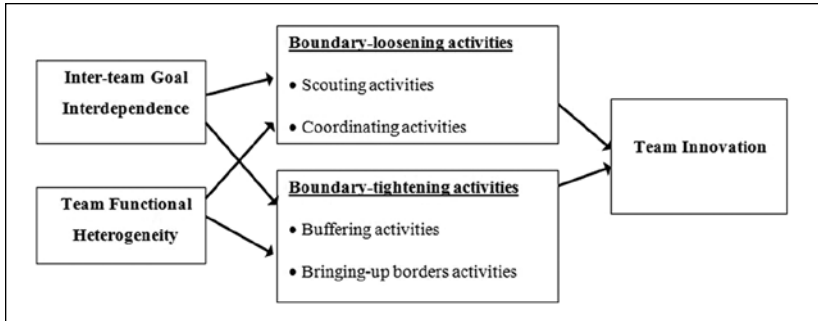


Figure 1. Research model.

The present study proposes a model that refers simultaneously to a team's boundary-tightening and -loosening activities, and examines them in a mediating model (see Figure 1). Based on the Social Identity Perspective (Ashforth & Mael, 1989; Brewer, 1996, 2000; Hogg, 2000; Tajfel & Turner, 1986), the model proposes that the boundary activities that one team adopts with respect to other teams are related to two structural variables: at the organizational level, inter-team goal interdependence; at the team level, team composition for functional heterogeneity. Furthermore, it proposes that a team's boundary activities are related to team innovation and that team boundary activities will mediate the relationships between the structural variables and team innovation. The focus here is on the interrelations among teams in the same organization.

Theoretical Background and Hypotheses

Research has typically referred to teams as isolated entities, separate from their organizations, in reference to processes as well as outcomes (Barrick et al., 2007). But teams do not operate in a vacuum; they function, for instance, through interdependent relationships with other organizational teams (van Knippenberg, 2003) to achieve their goals. As mentioned, these teams have to obtain necessary information, products, or services from other teams, which must be trusted to take their interests into consideration and constructively resolve potential conflicts of interest. The dearth of research on work units as open systems and on related boundary issues may at least partly be attributed to the fact that until quite recently work teams have largely been an integral part of the traditional structures of organizations. These hierarchical structures, whether functional or multidivisional in form,

have insulated teams from exposure to their external environments (Yan & Louis, 1999) by appointing special liaisons or departments within the organization responsible for boundary activities.

Only in the 1980s did team research begin to adopt an external viewpoint, which suggested that teams that managed their external boundaries and obtained crucial resources from their surroundings demonstrated better results than teams that managed only their own internal boundaries (Ancona & Caldwell, 1998; Sundstrom, de Meuse, & Futrell, 1990; Yan & Louis, 1999). On the basis of earlier research (Allen, 1977; Katz & Tushman, 1979; Malone, 1987), Ancona and Caldwell (1992) classified team's activities that focused outwardly toward other organizational departments in four distinct types: ambassadorial, scouting, guarding, and integrating. The approach of Ancona and Caldwell, later adopted by other scholars (Choi, 2002; Marrone, 2010), associates team boundary activities with outward activities, aimed at loosening the team boundary. These activities capture interactions *across* team boundaries to entities in the surroundings. By this approach, inward boundary-tightening activities through systematic monitoring and regulation of the inner system run counter to boundary-loosening processes (Marrone, 2010). More recently, Yan and Louis (1999) introduced another approach to the concept of boundary activity, later elaborated by other scholars (Drach-Zahavy & Somech, 2010; Faraj & Yan, 2009). These authors suggest that teams engage in *boundary-loosening activities* and *boundary-tightening activities* to delineate and maintain boundaries and to manage interactions across them. The former are intended to gain resources and information and to manage relationships with external stakeholders, the latter to protect team resources from leaking into the environment, as well as to strengthen the feeling of belonging and to build a distinct team identity to cultivate sustained commitment of their members to their common task (Druskat & Wheeler, 2003; Faraj & Yan, 2009).

Hence, a team's essential mission is to create its boundaries that are sufficiently permeable to allow resources and information in, yet sufficiently impermeable to prevent any doubt as to who belongs to the team and whether these members are accountable for the team's collective outputs (Faraj & Yan, 2009). In line with this thinking, we define a team's boundary activities as processes necessary for a team to carry out the task at hand, and directed at external agents in its focal environment to obtain resources and promote and protect itself (Drach-Zahavy & Somech, 2010; Yan & Louis, 1999). In that way, boundary activities should refer to all the activities that a team engages in to manage its boundaries. Team's boundaries may be crossed between the team and its organization, between teams within the same organization, and between the team and its external environment. Prior studies

(e.g., Ancona & Caldwell, 1998) looked into broad external targets, such as top management, suppliers, competitors, or customers. Of course, some decisions had to be made about which entities to focus on. Including simultaneously different external targets does not allow identification of the unique boundary activity the team exhibits toward each target; therefore, in the present study the decision was to focus on the interrelations among teams in the same organization.

Based on this holistic approach, a typology suggested by Drach-Zahavy and Somech (2010) is adopted here. It represents the full scope of boundary activities previously identified in the literature (Ancona & Caldwell, 1998; Yan & Louis, 1999). These are two boundary-loosening activities—scouting and coordinating—and two boundary-tightening activities—buffering and bringing-up borders. *Scouting* involves activities by a focal team to reach out to its environment for important resources and support (Faraj & Yan, 2009), for example, scanning for ideas, seeking support, and accessing information and other resources. These activities encompass many aspects of mapping, information gathering, and scanning, all aimed at enhancing the team's capability (Amedore & Knoff, 1993; Ancona & Caldwell, 1992, 1998; Druskat & Wheeler, 2003; Edmondson, 1999; Yan & Louis, 1999). *Coordinating* means the team's coordinating, collaborating, and negotiating with outsiders on work objectives and getting feedback from them. Ancona and Caldwell (1992), for example, describe a team's efforts to negotiate delivery deadlines with other teams. Other examples are coordinating technical or design issues as well as obtaining feedback on the team's functioning and performance. These activities, which promote working in greater harmony with other teams in the organization, often fill many of the gaps opened by formal integrating systems (Marrone, 2010; Somech & Drach-Zahavy, 2013).

As for the two boundary-tightening activities—buffering and bringing-up borders—*Buffering* stands for a team's insulating of the system from unwanted exposure to the environment: preventing disturbances and uncertainties, negative input caused by undesired access to the team boundaries, and leakage of valuable resources (Faraj & Yan, 2009; Yan & Louis, 1999). *Bringing-up borders* entails enlisting team members' attachment and energies as well as focusing them on accomplishing the common task. It involves creating a coherent unit for team members despite the diversity of their backgrounds; consequently, the boundaries that emerge between the team and its environment tend to be quite natural (Druskat & Wheeler, 2003; Yan & Louis, 1999). According to Yan and Louis (1999),

The task of bringing up work unit boundaries is accomplished through two related pursuits. The first is to attract the members' attention, energy, and

resources toward the unit's mission and tasks, thus developing a centripetal force; the second is to create a supportive climate in which a shared sense of uniqueness and distinctive character of the work unit emerges. (p. 33)

Examples of bringing-up borders are inviting team members to participate in developing the vision, encouraging support, and helping behaviors and developing a sense of safety and respect. These activities strengthen the team's distinction from other teams.

Antecedents of Boundary Activity

The first stage of the proposed model is to identify antecedents of boundary activities. The model proposes that two structural variables—inter-team goal interdependence and team composition of functional heterogeneity—are related to the extent to which a team engages in boundary loosening (scouting and coordinating) and/or in boundary tightening (buffering and bringing-up borders). These structural variables are, of course, only a subset of numerous variables that can be studied. Nevertheless, each of them is among the variables most studied in team literature and the hypotheses developed here are intended to offer insights into how inter- and intra-team dynamics can be related to the team's boundary activities.

The social identity approach (Ashforth & Mael, 1989; Brewer, 1996, 2000; Hogg, 2000; Tajfel & Turner, 1986) may provide the logic for the influence of structural variables on team activity. It holds that a potentially important and positively valued component of people's identities derives from their group memberships and that these positively valued *social identities* are maintained primarily by means of social comparisons of in-group members with relevant out-groups (Richter, West, van Dick, & Dawson, 2006). This tendency can be explained through a self-categorization that accentuates attitudinal, emotional, and behavioral similarity to the group prototype—one's representation of the features that best define the in-group in the salient social comparative context (Hogg & Hains, 1996).

From this angle, team-level identities are often prevalent in members' interactions within the organization due to physical proximity, interpersonal similarity, and so on (Ashforth & Mael, 1989). However, it is incorrect to claim that the social identity approach assumes that organizational identity cannot evolve together with team identity. Several important conditions are identified by this approach to lessen both favoritism of the in-group and derogatory behaviors and attitudes to the out-group (Bartel, 2001; Hogg & Terry, 2000; van Knippenberg, 2003). Structural variables, such as inter-team goal interdependence or team composition, can enhance an individual's

loyalty to the organization and reduce the importance of his or her primary loyalty to the team (Kramer, 1991).

Inter-team goal interdependence is the extent to which a team perceives that its goals and/or feedback it has received are congruent with the organization's goals (Saavedra, Earley, & Van Dyne, 1993). It may be seen as a continuum from slight interdependence, with no apparent common goal or need for cooperation to achieve the teams' goals, to abundant interdependence, with a well-defined common goal that cannot be attained without inter-team work and cooperation (Campion et al., 1996). Scholars have argued that high interdependent goals require high levels of external interaction to succeed (Hoegl, Weinkauff, & Gemuenden, 2004), so teams need to loosen their boundaries to allow exchange of resources with other teams. For example, to achieve their common goals, teams must coordinate adherence to project schedules or exchange of task-relevant information. Coordinating facilitates exchanges with specialists from other teams, through which new perspectives and alternative ideas enter the team (Hoegl et al., 2004; Joshi et al., 2009; Sethi, 2000; Tsai, 2000). Mathieu, Marks, and Zaccaro (2001) illustrate this by describing how firefighter and ambulance teams have to closely synchronize their efforts to extract injured motorists from a crash scene quickly and safely. Higher levels of inter-team goal interdependence will require teams to loosen their boundaries so as to gather and share knowledge, expertise, and resources, that is, to invest more efforts in coordinating and scouting (Joshi et al., 2009; Marrone, 2010).

Under low inter-team goal interdependence, teams will need to invest less in coordinating and scouting with other teams (Barrick et al., 2007). Instead, each team must strive to reach its own goals. It has no need to loosen its boundaries for interactions with other teams; it must tighten them to signal to team members to invest their efforts within to accomplish the team's goals (Drach-Zahavy & Somech, 2010).

The social identity approach also reinforces the logic that inter-team goal interdependence is a relevant structural variable for understanding team boundary activities. It suggests that high goal interdependence (superordinate goals) reduces bias and conflict, which, in turn, transforms the team members' self-perceptions from *us versus them* to the more inclusive *we*. In keeping with the notion of *community of fate* (Brewer, 2000), high inter-team goal interdependence creates greater coordination in teams than does low goal interdependence (van Vijfeijken, Kleingeld, van Tuijl, Algera, & Thierry, 2006).

Hypothesis 1a: There is a positive relationship between inter-team goal interdependence and scouting and coordinating.

Hypothesis 1b: There is a negative relationship between goal interdependence and buffering and bringing-up borders.

Team functional heterogeneity is defined as “the diversity of organizational roles embodied in the team” (Jackson, 1992, p. 353). Functionally heterogeneous teams assemble people from different disciplines and functions who have pertinent expertise in the proposed course of action (Earley & Mosakowski, 2000). When diverse professionals are grouped together as a cross-functional team, that team will be characterized as possessing increased functional heterogeneity. Functional heterogeneity stimulates communication with individuals from outside the team and has been associated with the volume of information exchanged, providing the team with broader informational resources and knowledge (Drach-Zahavy & Somech, 2010). Diversity in members’ experience, expertise, and previous membership on other teams can promote a team’s tendency to loosen its boundaries and connect with other teams, an asset that is not available to homogeneous teams (Arrow & McGrath, 1995). Therefore, heterogeneous teams are more likely to have more diverse contacts outside the group boundary that will facilitate scouting and coordinating (Choi, 2002).

This logic agrees with the social identity approach. Ashforth and Mael (1989) argued that members of homogeneous teams are more likely to view themselves as the in-group, hence to categorize those outside the team boundary as out-groups (Ashforth & Mael, 1989). In effect, when teams are homogeneous, they tend to produce perceptual boundaries and team favoritism that unite them, but also distinguish them from other teams (Joshi & Jackson, 2003). By contrast, the tendency in heterogeneous teams is for members to perceive their peers and the team itself less as an in-group, so that favoritism based on team membership is less predominant (Joshi, 2006; Joshi & Jackson, 2003). These members have a more limited sense of belonging to their team, due to their different disciplines, roles, and positions. Not surprisingly, members in heterogeneous teams might even prefer to associate with members of other teams that speak the same professional language. This tendency will result in their investing more time and energy in loosening team boundaries, namely, in scouting and coordinating (Choi, 2002); functionally homogeneous teams will invest their energy and efforts in tightening boundaries through systematic monitoring and regulating the internal system—namely, in bringing-up borders and buffering.

Hypothesis 2a: There is a positive relationship between team functional heterogeneity and scouting and coordinating activities.

Hypothesis 2b: There is a negative relationship between functional heterogeneity and buffering and bringing-up borders.

Team Boundary Activities and Innovation

Innovation has been highlighted as a core competence for contemporary organizations to maintain or enhance effectiveness in rapidly changing and challenging environments (Choi & Chang, 2009), and it seems to be relevant to team boundary activities (J. M. Howell & Shea, 2006). Team innovation is defined as “the intentional introduction and application within a team, of ideas, processes, products or procedures new to the team, designed to significantly benefit the individual, the team, the organization, or wider society” (West & Wallace, 1991, p. 303). Thus, innovation encompasses creativity (idea generation) as well as implementation (De Dreu & West, 2001; George, 2007; Shalley, Zhou, & Oldham, 2004). According to the present model, the four boundary activities can be linked to team innovation.

There are logical reasons to expect a positive relation between the boundary-loosening activities of scouting and coordinating, directed at crossing the team’s boundary outward, and innovation. The importance of external communication to successful innovation is well documented in the literature (J. M. Howell & Shea, 2006; Hülsheger, Anderson, & Salgado, 2009). Taking a social networks perspective, Perry-Smith and Shalley (2003) underlined the importance of external relations with people outside one’s own team. Interactions with other teams enhance the likelihood of obtaining new knowledge and revealing new perspectives, which spark the development of new ideas or the adoption of new ways of doing things. Previous research has demonstrated that information acquired through boundary spanning is a powerful resource for influencing organizational decisions and improving team decisions and innovation (Ancona & Caldwell, 1992; Druskat & Wheeler, 2003; J. M. Howell & Shea, 2006). For example, Allen (1971, 1977) found that high performing research and development teams communicated more frequently with others outside the team than did low performing teams. Similarly, J. M. Howell and Shea (2006), who also examined new-product development teams, found a positive link between scouting and coordinating activities and team innovation. According to these researchers, scouting activities enable the team to search other teams for information, to build support for the new product, and to influence the demands of others; interacting with other teams through coordinating activities, teams can exchange ideas, discuss divergent viewpoints, and integrate and evaluate them to create high-quality products or to suggest innovative procedures (Hülsheger et al., 2009).

Other studies, conversely, point to the importance of activities that tighten a team's boundaries, so as to boost members' sustained commitment to the task at hand (Ancona & Caldwell, 2007; Faraj & Yan, 2009). In their meta-analysis, Hülshager et al. (2009) found that, among other factors, both external and internal communications are strongly associated with team innovation. We argue that, to promote innovation, a team should not only loosen and cross the boundaries to the external environment, but also tighten its boundaries. Given that team innovation includes idea generation as well as idea implementation, these inward activities seem especially important. To foster innovation, team members need to tighten boundaries to monitor each other, help each other out, and provide constructive feedback (Faraj & Yan, 2009). Buffering enables teams to block external disturbances and prevent leakage of information or other resources needed to execute the innovative product (Scott, 1998). However, buffering the team from the demands of the environment may not be enough if there is no complementary bringing-up of borders to create a unique sense of belonging and team identity. These activities are crucial for attracting and concentrating members' attention on their central task of innovation (Faraj & Yan, 2009). Bringing-up borders by creating an atmosphere of support and mutual care allows teams to preserve the resources and apply them to developing team innovation (Yan & Louis, 1999).

Hypothesis 3: Boundary activities (scouting, coordinating, buffering, and bringing-up borders) are positively related to team innovation.

Mediation Model

Finally, the model proposes complete mediation of team boundary activities in the relationship of structural variables to team innovation. In the traditional inputs-process-outputs (I-P-O) framework for teams (Hackman, 1987), processes are the linking variables between inputs and outcomes. In this framework, inputs are the structural variables of the constellation of individual characteristics and resources on multiple levels (individual, team, organization). Processes are activities that team members engage in, combining their resources to resolve (or fail to resolve) task demands. Processes thus mediate the translation of inputs into outcomes. Specifically, the present model suggests that team structural variables (inter-team goal interdependence and functional heterogeneity) will promote team boundary activities, which in turn will enhance team innovation. Hackman and Morris (1975) noted that the team's design is one of the most potent determinants of what constitutes an effective process to promote team performance and innovation.

Accordingly, we suggest that the degree of inter-team goal interdependence and functional heterogeneity shape the pattern of teams' boundary activities, and this pattern of boundary activities, in turn, will determine the level of team innovation.

Hypothesis 4: Boundary activities (scouting, coordinating, buffering, and bringing-up borders) mediate the relationship of the structural variables of inter-team goal interdependence and functional heterogeneity to team innovation.

Method

Sample and Procedure

Sixty product development teams were recruited from different high-technology companies in Israel, specializing in telecommunications, computer software, computer hardware, and semiconductors. These teams are made up of personnel from different functional departments to support a new product's design, development, and transition to production. Each team leader was evaluated in a pre-assessment interview to ensure that each team provided an appropriate setting for this study. All interviewed team leaders stated that the members of their teams interacted regularly to achieve shared goals. Team members also depended on one another for knowledge and exerted joint efforts by means of several permanent arrangements, such as scheduled staff meetings, "brown bag" lunch meetings, and joint refresher workshops. A 60% response rate (but not less than four members) was the criterion for a team to be included in the final sample.

Data were collected from multiple sources at two time points. The questionnaire administered to team members covered scales on inter-team goal interdependence, boundary activities and demographic information. These measures were aggregated to the team level of analysis. However, to control for potential common source bias among the variables rated by team members, the criss-cross method was used (J. M. Howell & Shea, 2006; Podsakoff & Organ, 1986). Members of each team were randomly split into two groups. The responses of one group were used to assess inter-team goal interdependence and the responses of the second group were used to assess team boundary activities. A month later, the team leader was asked to evaluate team innovation and to provide demographic information regarding his or her team members.

Team size ranged from 6 to 15 members, with an average of 8.2 ($SD = 0.97$). These teams consisted of 56 engineers, 29 developers, 7 graphic designers, 10

technical leaders, 21 designers, 21 architects, 13 software testers, 9 quality assurance experts, 15 programmers, and 15 mentors. The period during which team members had been working together ranged from 1 to 7.9 years, with an average of 3.1 years ($SD = 1.7$). The sample consisted of 196 employees (82% men) and their 60 corresponding team leaders and unit managers. The participants' average age was 33.19 years ($SD = 3.46$), with an average time on the job of 5.7 years ($SD = 4.28$). They had at least a bachelor's degree, 39% in computer sciences and the rest in electronics; 20% of the participants had a master's degree. Analysis of variance of the team averages of the demographic variables confirmed no statistically significant differences across teams in terms of gender, age, job, tenure, or education. In addition, none of the demographic variables predicted a significant portion of the variance in team boundary activities or team innovation. Hence, these demographic variables were not included in subsequent analyses to test the hypotheses.

Measures

Team functional heterogeneity. Team functional heterogeneity was defined as the diversity of organizational roles embodied in the team (Jackson, 1992). Information for this measure was provided by each team's leader; therefore, functional heterogeneity was based on the actual composition of the team. Functional heterogeneity was measured by the diversity index recommended by Blau (1977) and used by Simons, Pelled, and Smith (1999), $1 - \sum \pi^2$, where π is the proportion of the total team that each functional category represents. This proportion is squared and summed over each category. The functional categories used were engineers, developers, graphic designers, technical leaders, designers, architects, software testers, quality assurance experts, programmers, and mentors.

Inter-team goal interdependence. Inter-team goal interdependence was assessed by the scale of intra-team goal interdependence developed by van der Vegt, van de Vliert, and Oosterhof (2003), and re-worded the items to the organizational level. Each team member rated the extent to which the teams in the organization are assigned collective goals and collective feedback ("Teams are informed about the goals they should attain as an organization"; $\alpha = .83$).

Boundary activities. Team members were asked to assess to what extent their team engaged in boundary activities. The questionnaire had four dimensions. The scales for assessing the three dimensions of scouting, coordinating, and buffering were based on the questionnaire developed by Ancona (1993). The

bringing-up borders scale was based on the questionnaire developed by Faraj and Yan (2009). The scouting subscale consists of 9 items ("Scans other teams for ideas," $\alpha = .84$), the coordinating subscale consists of 5 items ("Coordinates activities with other teams," $\alpha = .83$), the buffering subscale consists of 8 items ("Protects the team from other teams' interference," $\alpha = .80$), and the bringing-up borders subscale consists of 4 items ("The team tries to create a clear sense of its identity and purpose," $\alpha = .83$).

The boundary activities subscales were analyzed using a series of confirmatory factor analyses to determine the appropriateness of proceeding with four separate subscales. Specifically, we compared a four-factor model with a two-factor model, as well as a single-factor model. The results indicated that the two-factor model provided a better fit than the two other models (two-factor model: normed fit index [NFI] = .98, comparative fit index [CFI] = .96, root mean square error of approximation [RMSEA] = .02; four-factor model: NFI = .80, CFI = .80, RMSEA = .16; and single-factor model: NFI = .88, CFI = .89, RMSEA = .10). These results provide additional support to the claim that treating boundary activity as a two-dimensional construct may be more appropriate than treating it as a four- or one-dimensional construct.

Team innovation. Team innovation was measured by a four-item scale adapted from West and Wallace (1991), and was evaluated by the team leader. The items reflected the extent to which, in the previous 6 months, the team had (a) initiated new procedures and methods, (b) developed innovative ways of accomplishing work targets/objectives, (c) developed new skills to foster innovation, and (d) developed new products. Following West and Wallace's instructions, a time frame of 6 months was provided, because it is easier for individuals to respond to questions about processes if provided with a specific time frame, rather than a more general question about the level of innovation. Furthermore, the leader's rating was used for this study because more objective measures of innovation at the team level were unavailable and, in any case, are incomparable across organizations; they are also recognized as problematic and subject to manipulation (Faraj & Yan, 2009).

Level of Analysis

In the present study, the unit of analysis was the team. All the hypotheses referred to the team and such study variables as boundary activities and inter-team goal interdependence were framed at the team level, and aggregated to team level. Rousseau (1985) claimed that aggregation is justified by both theoretical and empirical arguments. From a theoretical standpoint, she advocated using composition theories, which describe different levels of the

functional similarities of constructs. One might expect team members to share their perceptions of goal interdependence and team activities in their work environment for several reasons. Shared norms and collective views and perceptions may be engendered and developed through the team members' shared tasks, clearly delineated team boundaries, frequent interaction, and long-standing status of the team (George, 1990; Jehn, Chadwick, & Thatcher, 1997).

Therefore, it was critical to demonstrate high within-team agreement to justify using the team average as an indicator of a team-level variable (r_{wg} ; James, Demaree, & Wolf, 1993). A value of .70 or above is suggested as a good amount of within-group interrater agreement (James et al., 1993). All scales exceeded this criterion (for inter-team goal interdependence, the averaged r_{wg} score was .93, ranging from .81 to .96; for boundary-loosening activities, the averaged r_{wg} score was .90, ranging from .84 to .93; and for boundary-tightening activities, the averaged r_{wg} score was .96, ranging from .90 to .99). The within-group agreement was further estimated by two measures: ICC(1) provides an estimate of the reliability of an individual respondent's rating, and ICC(2) estimates the reliability of mean differences across teams (between-group variance). The following ICC(1) and ICC(2) values were obtained: inter-team goal interdependence, .33 and .82; boundary-loosening activities, .20 and .80; and boundary-tightening activities, .29 and .84. As indicated by James et al. (1993), ICC(1) generally ranges from 0 to .50 with a median of .12. However, no definite guidelines exist to determine acceptable values for ICC(2). The present values were comparable with the median or recommended ICC values; accordingly, it was concluded that aggregation was justified for these variables.

Results

Table 1 presents the means, standard deviations, and intercorrelations matrix for all key variables included in the analysis.

Model Comparisons and Overall Model Fit

To test the study model, we conducted structural equation modeling (SEM) using the AMOS 5.0 program. In their article, Blakely, Srivastava, and Moorman (2005) argued that structural equation modeling is a more appropriate technique than multiple regression for testing mediation. Structural equation modeling is a confirmatory approach in which the model being tested represents the hypothesized relationships among an initial variable, a mediator, and an outcome variable. Those relationships are tested simultaneously.

Table 1. Means, Standard Deviations, and Correlation Matrix of Study's Variables ($N = 60$).

	<i>M</i>	<i>SD</i>	1	2	3	4	5
1. Inter-team goal interdependence	3.78	0.42	1.00				
2. Functional heterogeneity	0.48	0.49	-.14	1.00			
3. Boundary-loosening activities	3.55	0.36	.44**	.26*	1.00		
4. Boundary-tightening activities	3.21	0.38	.11	-.25*	-.02	1.00	
5. Team innovation	3.94	0.60	.28*	.26*	.23*	.45**	1.00

* $p < .05$. ** $p < .01$.

To support either full or partial mediation, all parameter estimates in the hypothesized model should be significant. For full mediation, the model should also acceptably capture the observed correlation matrix, per goodness of fit tests (James, Mulaik & Brett, 2006).

In the present study, the full mediating model (Model 1) describes the relationship of inter-team goal interdependence and team functional heterogeneity (initial variables) to boundary-loosening activities and to boundary-tightening activities (mediating variables), and that of the mediating variables to team innovation (outcome variable). For supporting the full mediation model (Model 1), we compared Model 1 with a partially mediated model (Model 2), which is the same model but with additional paths—from the initial variables (inter-team goal interdependence and team functional heterogeneity) to the outcome variable (team innovation).

The results of the SEM analyses indicated that these models were not significantly different, $\Delta\chi^2(2) = 0.42$, $p > .05$, indicating that the addition of these paths for partial mediation did not add significantly to the model. Model 1 fit the data well ($\chi^2 = 7.43$, $df = 11$, $p > .05$, $NFI = .99$, $CFI = .99$, $RMSEA = .001$), providing overall support for the mediated model. The results of the mediating model are presented in Figure 2.

Tests of Hypotheses

Hypothesis 1a, which proposed a positive relationship between inter-team goal interdependence and scouting and coordinating activities (boundary-loosening activities) was supported. As shown in Figure 2, the analysis revealed that inter-team goal interdependence was positively and significantly

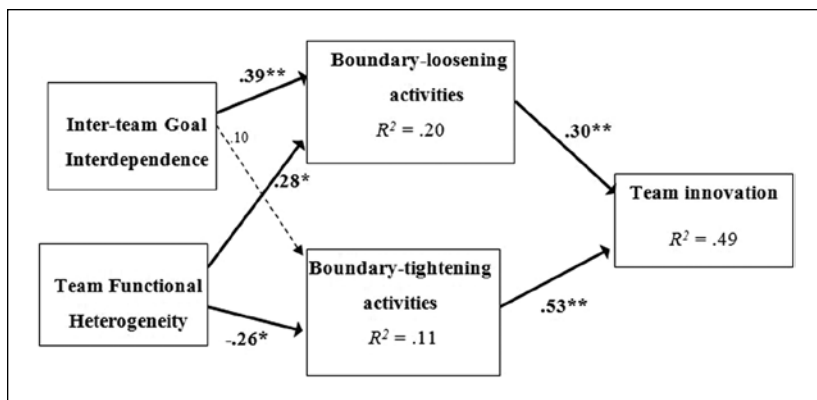


Figure 2. Estimated path coefficients for the model.

Note. Standardized parameter estimates for the theoretical model.

Nonsignificant relationship: * $p < .05$. ** $p < .001$.

related to boundary-loosening activities ($\beta = .39$, $p < .001$). However, Hypothesis 1b, which proposed a negative relationship between goal interdependence and buffering and bringing-up borders (boundary-tightening activities) was not supported. The path between inter-team goal interdependence and boundary-tightening activities was not significant.

Hypothesis 2a suggested a positive relationship between team functional heterogeneity and scouting, and coordinating activities (boundary-loosening activities), while Hypothesis 2a suggested a negative relationship between functional heterogeneity and buffering and bringing-up borders (boundary-tightening activities). The results, which indicated a positive and significant relationship with boundary-loosening activities ($\beta = .28$, $p < .05$), and a negative and significant relationship with boundary-tightening activities ($\beta = -.26$, $p < .05$) supported the hypotheses.

Hypothesis 3 proposed that boundary activities (scouting, coordinating, buffering, and bringing-up borders) are positively related to team innovation. The analysis, which indicated a positive and significant relation of boundary-loosening activities with team innovation ($\beta = .30$, $p < .001$) as well as between boundary-tightening activities and team innovation ($\beta = .53$, $p < .001$), supported the hypothesis.

Finally, as described above, the results of the SEM analyses provided overall support for Hypothesis 4, which suggested that boundary activities mediate the relationship between the structural variables of inter-team goal interdependence and functional heterogeneity and team innovation.

Discussion

The findings presented herein provide support for the notion that team boundary activities serve as mediators in the relationship between the team's structural characteristics and team outcome of innovation. Overall, these findings emphasize the role of boundary activities in promoting team innovation, and contribute to the team literature in several ways.

First, regarding team boundary activities' typology, we referred to four distinctive activities: scouting, coordinating, bringing-up borders, and buffering. However, the results supported a two-dimensional classification, labeled here as boundary-loosening activities and boundary-tightening activities. The former emphasizes the extent to which a team invests in interacting with other teams in the organization. These outward-facing boundary activities allow the team members to gain access to strategic organizational information and important resources (J. M. Howell & Shea, 2006), while the boundary-tightening activities are inwardly focused activities that aim to maintain team identity by sharpening the team boundary and enhancing team cohesiveness (Faraj & Yan, 2009). These findings, which are in congruence with previous studies (Faraj & Yan, 2009; Yan & Louis, 1999), provide additional validation for the boundary activities framework that distinguishes between deliberate strategies for loosening boundaries and strategies aiming to tighten boundaries through systematic monitoring and regulation of the inner system (Yan & Louis, 1999).

Druskat and Wheeler (2003), writing in the same vein, demonstrated that effective team leaders are those that cross boundaries, back and forth, to seek vital information, build important relationships, and persuade their team and other entities to support one another and inspire their team to attain success. It seems that, although these activities might compete for limited resources within a team, they may maintain synergistic relationships through mutual reinforcement (Choi, 2002). Therefore, this holistic perspective is important because it enables to move from an either/or approach to a both/and approach, for better understanding of the dynamic of teamwork (Somech, 2006).

Second, our model's results demonstrated a positive association between boundary-loosening activities and team innovation as well as between boundary-tightening activities and team innovation. Although until now several researchers (Ancona, 1993; J. M. Howell & Shea, 2006) have highlighted the importance of activities directed toward agents and players external to the focal team for promoting team innovation, findings from this study support the notion that for promoting innovation, it is necessary for a team to invest time and effort simultaneously in outward as well as in inward boundary activities. Boundary-loosening activities are critical for enhancing

innovation, since they provide opportunities for gaining needed resources, for exchanging ideas, and for learning (Johnson & Chang, 2000). Boundary-tightening activities might allow a team to build a team identity and cohesiveness, which are important for motivating members to incorporate their efforts within the team for successfully integrating information from outside sources, which is essential for promoting innovation (Ancona & Caldwell, 2007). Gersick (1988) found that teams shift their operating procedures over time. She suggests that teams may deal with initial information from the environment in isolation and then emerge to obtain further feedback and information from outsiders. Tushman (1979), by contrast, reported a two-step communication process in high performing teams, featuring communication that starts initially with gathering information from outside the team, and then interpreting that information and transmitting it inwardly to the team. Overall, the present results might suggest that for enhancing team innovation, teams need deliberate strategies for reaching out to the external world as well as strategies aiming to motivate members to pull together and coalesce.

Third, the findings showed that structural variables are related to the extent to which teams engage in both boundary-loosening activities and boundary-tightening activities. Specifically, the results indicated a positive link between inter-team goal interdependence and boundary-loosening activities, but no significant link with boundary-tightening activities. In other words, high inter-team goal interdependence encourages teams to invest in outward boundary activities of communication and coordination with their counterparts for their mutual benefit (Hülshager et al., 2009; Tjosvold, Tang, & West, 2004). This finding is in congruence with the resource dependence perspectives (Astley & Sachdeva, 1984; McCann & Ferry, 1979), which suggest that a common purpose among business units creates a shared language and provides incentives to invest in outward boundary activities for developing inter-unit linkages (Tsai, 2000). For example, Joshi et al. (2009), who conducted an interview study, provided rich illustrations of the role of interdependence in shaping team boundary spanning.

Regarding functional heterogeneity, the results demonstrated a positive association with boundary-loosening activities and a negative association with boundary-tightening activities. These results support previous arguments, which claim that functionally heterogeneous teams are more likely to have contacts of greater number and diversity outside the team boundary, namely, engaging more in boundary-loosening activities. However, at the same time, they might neglect those inward activities of boundary-tightening activities aimed at building intra-group cohesiveness and team identity (Choi, 2002).

Finally, this study addresses the call for research in identifying mediating variables in team innovation (Eisenbeiss, van Knippenberg, & Boerner, 2008) and demonstrates that boundary activities mediate the relationship between the structural variables of inter-team goal interdependence and functional heterogeneity and team innovation. Specifically, we found that boundary-loosening activities serve as a mediator between the two structural variables and team innovation, while boundary-tightening activities mediate only between functional heterogeneity and team innovation. These findings support the classic model of I-P-O, which emphasizes that team boundary activities serve as a vehicle whereby the structural variables of inter-team goal interdependence and team functional heterogeneity enhance team innovation. Accordingly, the present results demonstrate that inter-team goal interdependence and functional heterogeneity “set the stage” for encouraging the team to invest in both boundary-loosening and boundary-tightening activities, which enable the team to reach out into its environment to obtain important resources and support as well as to attract members’ energies and attachment for focusing them on accomplishing the task to maximize their innovation (Yan & Louis, 1999).

Limitations and Future Research

Finally, the results of this study are obviously limited by its operations. First, for example, since the data may be subject to bias, as they are largely self-reported, respondents might anticipate what the researchers seek to prove and thus respond in ways that confirm these supposed hypotheses. Alternatively, they might hold some implicit theory concerning the relationships among the variables and set out to confirm that theory. Recent research, however, suggests that people often accurately perceive their social environment and that self-reported data are not as limited as previously believed (Alper, Tjosvold, & Law, 1998). Furthermore, because we used the criss-cross method for assessing inter-team goal interdependence and team boundary activities and the team leader evaluated the team’s innovation, common method variance may not be a serious problem in our data (J. M. Howell & Shea, 2006; Podsakoff & Organ, 1986). Second, our sample size is rather small so caution is needed in generalizing conclusions. The results need to be replicated and confirmed in a future study applying SEM models to a larger sample.

Third, the issue of causality arises from the present study’s design. The nature of the relationship between the team boundary activities and team innovation is quite difficult to determine. For example, one may ask, in regard to various team activities, “Do they determine innovation or are they a consequence of it?” Evidence of causal links between the structural variables of

goal interdependence and heterogeneity were not provided unequivocally by the data. It is highly possible that teams that produce positive processes will initially attract and, ultimately, preserve, a more heterogeneous team of members who will be inspired to set higher independent goals. Furthermore, the relationships might be of a spiral rather than a linear nature, namely, where structure, process, and outcomes are interrelated and mutually nurturing. To further explore the nature of these relationships, longitudinal studies using more objective criteria are clearly required.

Fourth, this study conceptualized boundary activity in a more holistic perspective by referring to two distinct categories of boundary activities: boundary-loosening activities and boundary-tightening activities, which refer to outwardly as well as inwardly facing boundary activities. Further research is needed to provide additional validation to this two-dimensional classification.

Finally, this study focuses on the two structural variables of inter-team goal interdependence and team functional heterogeneity as antecedents. These are just examples of the influences of context variables on team boundary activities and, in turn, on team innovation. Identifying other variables is especially important, because in the present study no variables were positively related to boundary-tightening activities. Future research should seek other antecedents embedded in the team context on various levels (e.g., Chen, Bliese, & Mathieu, 2005; Kozlowski & Klein, 2000). For example, we found a positive link between inter-team goal interdependence and boundary loosening, but no significant link with boundary tightening. However, intra-team variables, such as intra-team goal interdependence, might be positively related to boundary tightening and future research may well examine this parameter at parallel levels of team, organization, and industry. Researchers could benefit thereby in terms of generalizing boundary activity theory from the organizational to the interorganizational level, while at the same time providing a basis for an explicit comparison of similarities and differences in the determinants of boundary activities across levels (Drach-Zahavy & Somech, 2010; Santos & Eisenhardt, 2005).

Managerial Implications

Finally, our study also carries managerial implications. Today, most organizational teams in the contemporary business environment may need to deal with conflicting demands and goals. Developing an understanding of how teams deal with other groups can have important implications for helping improve the outcomes of these teams (Ancona & Caldwell, 2007). Our results emphasize the importance of developing a full range of boundary activities: outward-facing boundary activities boundary (loosening activities), which

help team members gain access to strategic organizational information and important resources, and simultaneously inwardly focused activities (boundary-tightening activities), which help team members maintain team identity and cohesiveness (Faraj & Yan, 2009; J. M. Howell & Shea, 2006). Team leaders should encourage team members to shift attention and allegiance back and forth from the team to its external environment (Druskat & Wheeler, 2003). Based on the results of this study, to promote a team's boundary-loosening activities, it is best to assemble members with high functional diversity and to create a work situation with high inter-team goal interdependence. The level of inter-team goal interdependence could be altered by the formulation of joint organizational objectives, and provision of organizational feedback. By creating the high inter-team goal interdependence, organizations may encourage teams to loosen their boundaries and stimulate cooperation among teams, which is an important function of modern team-based organizations (Barrick et al., 2007).

In conclusion, this study highlights the importance of more fully understanding the role of team boundary activities in promoting innovation in team-based organizations. The results provide empirical support for the notion that future research should take a broader perspective for understanding the phenomenon, by referring simultaneously to boundary-loosening and boundary-tightening activities. Furthermore, the study illustrates the mediating role of team boundary activities in the relation of the structural variables of inter-team goal interdependence and functional heterogeneity to team innovation.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

References

- Allen, T. J. (1971). Communication networks in R&D laboratories. *R&D Management*, 1, 14-21.
- Allen, T. J. (1977). *Managing the flow of technology: Technology transfer and the dissemination of technological information within the R&D organization*. Cambridge, MA: MIT Press.
- Alper, S., Tjosvold, D., & Law, K. S. (1998). Interdependence and controversy in group decision making: Antecedents to effective self-managing teams. *Organizational Behavior and Human Decision Processes*, 74, 33-52.

- Amedore, G. H., & Knoff, H. M. (1993). Boundary spanning activities and the multidisciplinary team process: Characteristics affecting school psychological consultation. *Journal of Educational and Psychological Consultation*, 4, 343-356.
- Ancona, D. G. (1993). The classic and the contemporary: A new blend of small group theory. In K. Murnighan (Ed.), *Social psychology in organizations: Advances in theory and research* (pp. 225-243). Englewood Cliffs, NJ: Prentice Hall.
- Ancona, D. G., & Caldwell, D. F. (1992). Demography and design: Predictors of new product team performance. *Organization Science*, 3, 321-341.
- Ancona, D. G., & Caldwell, D. F. (1998). Rethinking coalition composition from the outside in. In D. H. Gruenfeld (Ed.), *Research on managing groups and coalitions composition* (pp. 21-37). Stamford, CT: JAI Press.
- Ancona, D. G., & Caldwell, D. F. (2007). Improving the performance of new product teams. *Research-Technology Management*, 50(5), 37-43.
- Arrow, H., & McGrath, J. E. (1995). Membership dynamics in groups at work: A theoretical framework. In B. M. Staw & L. L. Cummings (Eds.), *Research in organizational behavior* (Vol. 17, pp. 373-411). Greenwich, CT: JAI Press.
- Ashforth, B. E., & Mael, F. (1989). Social identity theory and the organization. *Academy of Management Review*, 14, 20-39.
- Astley, W. G., & Sachdeva, P. S. (1984). Structural sources of intraorganizational power: A theoretical synthesis. *Academy of Management Review*, 9, 104-113.
- Barrick, M. R., Bradley, B. H., Kristof-Brown, A. L., & Colbert, A. E. (2007). The moderating role of top management team interdependence: Implications for real teams and working groups. *Academy of Management Journal*, 50, 544-577.
- Bartel, C. A. (2001). Social comparisons in boundary-spanning work: Effects of community outreach on members' organizational identity and identification. *Administrative Science Quarterly*, 46, 379-413.
- Blakely, G. L., Srivastava, A., & Moorman, R. H. (2005). The effects of nationality, work role centrality, and work locus of control on role definitions of OCB. *Journal of Leadership & Organizational Studies*, 12, 103-117.
- Blau, P. M. (1977). *Inequity and heterogeneity*. New York, NY: Free Press.
- Brewer, M. B. (1996). When contact is not enough: Social identity and intergroup cooperation. *International Journal of Intercultural Relations*, 20, 291-304.
- Brewer, M. B. (2000). Social identity theory and change in intergroup relations. In D. Capozza & R. Brown (Eds.), *Social identity processes* (pp. 117-132). London, England: Sage.
- Campion, M. A., Papper, E. M., & Medsker, G. J. (1996). Relations between work team characteristics and effectiveness: A replication and extension. *Personnel Psychology*, 49, 429-452.
- Chen, G., Bliese, P. D., & Mathieu, J. E. (2005). Conceptual framework and statistical procedures for delineating and testing multilevel theories of homology. *Organizational Research Methods*, 8, 375-409.
- Choi, J. N. (2002). External activities and team effectiveness: Review and theoretical development. *Small Group Research*, 33, 181-192.

- Choi, J. N., & Chang, J. Y. (2009). Innovation implementation in the public sector: An integration of institutional and collective dynamics. *Journal of Applied Psychology, 94*, 245-253.
- De Dreu, C. K. W., & West, M. A. (2001). Minority dissent and team innovation: The importance of participation in decision making. *Journal of Applied Psychology, 86*, 1191-1201.
- Drach-Zahavy, A., & Somech, A. (2010). From an intrateam to an interteam perspective of effectiveness: The role of interdependence and boundary activities. *Small Group Research, 41*, 143-174.
- Druskat, V. U., & Wheeler, J. V. (2003). Managing from the boundary: The effective leadership of self-managing work teams. *Academy of Management Journal, 46*, 435-457.
- Earley, P. C., & Mosakowski, E. M. (2000). Creating hybrid team cultures: An empirical test of international team functioning. *Academy of Management Journal, 43*, 26-49.
- Edmondson, A. (1999). Psychological safety and learning behavior in work teams. *Administrative Science Quarterly, 44*, 350-383.
- Eisenbeiss, S. A., van Knippenberg, D., & Boerner, S. (2008). Transformational leadership and team innovation: Integrating team climate principles. *Journal of Applied Psychology, 93*, 1438-1446.
- Faraj, S., & Yan, A. (2009). Boundary work in knowledge teams. *Journal of Applied Psychology, 94*, 604-617.
- George, J. M. (1990). Personality, affect, and behavior in groups. *Journal of Applied Psychology, 75*, 107-116.
- George, J. M. (2007). Creativity in organization. *The Academy of Management Annals, 1*, 439-477.
- Gersick, C. J. G. (1988). Time and transition in work teams: Toward a new model of group development. *Academy of Management Journal, 31*, 9-41.
- Hackman, J. R. (1987). The design of work teams. In J. Lorsch (Ed.), *Handbook of organizational behavior* (pp. 315-342). New York, NY: Prentice Hall.
- Hackman, J. R., & Morris, C. G. (1975). Group tasks, group interaction process, and group performance effectiveness: A review and proposed integration. In L. Berkowitz (Ed.), *Advances in experimental social psychology* (pp. 413-429), Academic Press, New York.
- Hoegl, M., Weinkauff, K., & Gemuenden, H. G. (2004). InterTEAM coordination, project commitment, and teamwork in multiteam R&D projects: A longitudinal study. *Organization Science, 15*, 38-55.
- Hogg, M. A. (2000). Social identity and social comparison. In J. Suls & L. Wheeler (Eds.), *Handbook of social comparison: Theory and research* (pp. 401-421). New York, NY: Kluwer Academic.
- Hogg, M. A., & Hains, S. C. (1996). Intergroup relations and group solidarity: Effects of group identification and social beliefs on depersonalized attraction. *Journal of Personality and Social Psychology, 70*, 295-309.
- Hogg, M. A., & Terry, D. J. (2000). Social identity and self-categorization processes in organizational contexts. *Academy of Management Review, 25*, 121-140.

- Howell, J. M., & Shea, C. M. (2006). Effects of champion behavior, team potency, and external communication activities on predicting team performance. *Group & Organization Management, 31*, 180-211.
- Howell, J. P., & Costley, D. L. (2006). *Understanding behaviors for effective leadership* (2nd ed.). Englewood Cliffs, NJ: Prentice Hall.
- Hülshager, U. R., Anderson, N., & Salgado, J. F. (2009). Team-level predictors of innovation at work: A comprehensive meta-analysis spanning three decades of research. *Journal of Applied Psychology, 94*, 1128-1145.
- Jackson, S. E. (1992). Team composition in organizational settings: Issues in managing an increasingly diverse workforce. In S. Worchel, W. Wood, & J. Simpson (Eds.), *Group process and productivity* (pp. 138-173). Newbury Park, CA: Sage.
- James, L. R., Demaree, R. G., & Wolf, G. (1993). An assessment of within-group interrater agreement. *Journal of Applied Psychology, 78*, 306-309.
- James, L. R., Mulaik, S. A., & Brett, J. M. (2006). A tale of two methods. *Organizational Research Methods, 9*, 233-244.
- Jehn, K., Chadwick, C., & Thatcher, S. M. B. (1997). To agree or not to agree: The effects of value congruence, individual demographic dissimilarity, and conflict on workgroup outcomes. *International Journal of Conflict Management, 8*, 287-305.
- Johnson, J. D., & Chang, H. J. (2000). Internal and external communication, boundary spanning, and innovation adaptation: An over-time comparison of three explanations of internal and external innovation communication in a new organizational form. *Journal of Business Communication, 37*, 238-263.
- Joshi, A. (2006). The influence of organizational demography on the networking behavior of teams. *Academy of Management Review, 31*, 583-595.
- Joshi, A., & Jackson, S. E. (2003). Managing workforce diversity to enhance cooperation in organizations. In M. A. West, D. Tjosvold, & K. Smith (Eds.), *International handbook of organizational teamwork and cooperative working* (pp. 277-296). New York, NY: Wiley.
- Joshi, A., Pandey, N., & Han, G. (2009). Bracketing team boundary spanning: An examination of task-based, team-level, and contextual antecedents. *Journal of Organizational Behavior, 30*, 731-759.
- Katz, R., & Tushman, M. (1979). Communication patterns, project performance and task characteristics: An empirical evaluation and integration in an R&D setting. *Organizational Behavior and Human Performance, 23*, 139-162.
- Kozlowski, S. W. J., & Klein, K. J. (2000). A multilevel approach to theory and research in organizations: Contextual, temporal, and emergent processes. In K. J. Klein & S. W. J. Kozlowski (Eds.), *Multilevel theory, research, and methods in organizations: Foundations, extensions, and new directions* (pp. 3-90). San Francisco, CA: Jossey-Bass.
- Kramer, R. (1991). Intergroup relations and organizational dilemmas. In L. L. Cummings & B. M. Staw (Eds.), *Research in organizational behavior* (Vol. 13, pp. 191-228). Greenwich, CT: JAI Press.

- Malone, T. W. (1987). *Modeling coordination in organizations and markets: Fulfilling the promise of the new organization*. Boston, MA: Harvard Business School Press.
- Marks, M., Mathieu, J., & Zaccaro, S. (2001). A temporally based framework and taxonomy of team processes. *Academy of Management Review*, 26, 356-376.
- Marrone, J. A. (2010). Team boundary spanning: A multilevel review of past research and proposals for the future. *Journal of Management*, 36, 911-940.
- Mathieu, J. E., Marks, M. A., & Zaccaro, S. J. (2001). Multiteam systems. In N. Anderson, D. S. Ones, H. K. Sinangil, & C. Viswesvaran (Eds.), *Handbook of industrial, work, and organizational psychology* (pp. 289-313). London, England: Sage.
- McCann, J. E., & Ferry, D. L. (1979). An approach for assessing and managing inter-unit interdependence. *Academy of Management Review*, 4, 113-119.
- Perry-Smith, J. E., & Shalley, C. E. (2003). The social side of creativity: A static and dynamic social network perspective. *Academy of Management Review*, 28, 89-107.
- Podsakoff, P. M., & Organ, D. W. (1986). Self-reports in organizational research: Problems and prospects. *Journal of Management*, 12, 531-544.
- Richter, A. W., West, M. A., van Dick, R., & Dawson, J. F. (2006). Boundary spanners' identification, intergroup contact, and effective intergroup relations. *Academy of Management Journal*, 49, 1252-1269.
- Rousseau, D. M. (1985). Issues of level in organizational research: Multi-level and cross-level perspectives. In B. M. Staw & L. L. Cummings (Eds.), *Research in organizational behavior* (pp. 1-37). Greenwich, CT: JAI Press.
- Saavedra, R., Earley, P. C., & Van Dyne, L. (1993). Complex interdependence in task-performing groups. *Journal of Applied Psychology*, 78, 61-72.
- Santos, F. M., & Eisenhardt, K. M. (2005). Organizational boundaries and theories of organization. *Organization Science*, 16, 491-508.
- Scott, W. R. (1998). *Organizations: Rational, natural, and open systems*. Englewood Cliffs, NJ: Prentice Hall.
- Sethi, R. (2000). Superordinate identity in cross-functional product development teams: Its antecedents and effect on new product performance. *Journal of Academy of Marketing Science*, 28, 330-341.
- Shalley, C. E., Zhou, J., & Oldham, G. R. (2004). The effects of personal and contextual characteristics on creativity: Where should we go from here? *Journal of Management*, 30, 933-958.
- Simons, T., Pelled, L. H., & Smith, K. A. (1999). Making use of difference: Diversity, debate, and decision comprehensiveness in top management teams. *Academy of Management Journal*, 6, 662-673.
- Somech, A. (2006). The effects of leadership style and team process on performance and innovation in functionally heterogeneous teams. *Journal of Management*, 32, 132-157.
- Somech, A., & Drach-Zahavy, A. (2013). Toward a conceptualizations of inter-team boundary activities: Identifying intra- and inter-team structural facilitators. In

- J. Langan Fox & C. L. Cooper (Eds.), *Boundary spanning in organizations: Network, influence, and conflict* (pp. 99-115). Routledge, New York: Taylor & Francis.
- Sundstrom, E., de Meuse, K. P., & Futrell, D. (1990). Work teams: Applications and effectiveness. *American Psychologist*, 45, 120-133.
- Tajfel, H., & Turner, J. (1986). The social identity theory of inter-group behavior. In S. Worchel & W. G. Austin (Eds.), *Psychology of intergroup relations* (pp. 518-538). Chicago, IL: Nelson.
- Tjosvold, D., Tang, M., & West, M. A. (2004). Reflexivity for team innovation in China: The contribution of goal interdependence. *Group & Organization Management*, 29, 540-559.
- Tsai, W. (2000). Social capital, strategic relatedness and the formation of intraorganizational linkages. *Strategic Management Journal*, 21, 925-939.
- Tushman, M. L. (1979). Work characteristics and solo unit communication structure: A contingency analysis. *Administrative Science Quarterly*, 24, 82-98.
- van der Vegt, G., van de Vliert, E., & Oosterhof, A. (2003). Informational dissimilarity and organizational citizenship behavior: The role of intrateam interdependence and team identification. *Academy of Management Journal*, 46, 715-727.
- van Knippenberg, D. (2003). Intergroup relations in organizations. In M. West, D. Tjosvold, & K. G. Smith (Eds.), *International handbook of organizational teamwork and cooperative working* (pp. 381-399). Chichester, UK: Wiley.
- van Vijfeijken, H., Kleingeld, A., van Tuijl, H., Algera, J. A., & Thierry, H. (2006). Interdependence and fit in team performance management. *Personnel Review*, 35, 98-117.
- West, M. A., & Wallace, M. (1991). Innovation in health care teams. *European Journal of Social Psychology*, 21, 303-315.
- Yan, A., & Louis, M. R. (1999). The migration of organizational functions to the work unit level: Buffering, spanning, and bringing up boundaries. *Human Relations*, 52, 25-47.

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